

Name: Solutions Student ID: _____

1. (10 points) Kelsey's Marshallian demand for x is given by

$$g_x(p_x, p_y, I) = \frac{I - 2p_x}{3p_y}$$

Her income is $I = 20$, and she faces prices $p_x = 4$ and $p_y = 1$. What is the income elasticity of demand for y ?

$$g_x = \frac{20 - 2(4)}{3(1)} = 4$$

$$e_{xI} = \frac{1}{3p_y} \cdot \frac{I}{g_x} = \frac{1}{3} \cdot \frac{20}{4} = \frac{5}{3}$$

$$s_x = \frac{p_x g_x}{I} = \frac{4(4)}{20} = \frac{4}{5}$$

$$s_y = 1 - s_x = \frac{1}{5}$$

$$s_x e_{xI} + s_y e_{yI} = 1$$

$$\frac{4}{5} \left(\frac{5}{3} \right) + \frac{1}{5} (e_{yI}) = 1$$

$$e_{yI} = 5 \left(1 - \frac{4}{3} \right) = \boxed{-\frac{5}{3}}$$

2. (10 points) Consider the following production function:

$$q = L^{1/3} K^{2/3}$$

Using factor prices v for capital and w for labor, solve for one of the contingent demands, either L^c or K^c .

Bonus (+1 point): Find both contingent demands.

$$RTS = \frac{\frac{1}{3} L^{-2/3} K^{2/3}}{\frac{2}{3} L^{1/3} K^{-1/3}} = \frac{K}{2L} = \frac{w}{v}$$

$$K = 2L \frac{w}{v}$$

$$q = L^{1/3} \left(2L \frac{w}{v} \right)^{2/3}$$

$$q = L \left(2 \frac{w}{v} \right)^{2/3}$$

$$\frac{q}{\left(2 \frac{w}{v} \right)^{2/3}} = L$$

$$\boxed{L^c = q \left(\frac{v}{2w} \right)^{2/3}}$$

$$K^c = 2 \left(q \left(\frac{v}{2w} \right)^{2/3} \right) \frac{w}{v}$$

$$\boxed{K^c = q \left(2 \frac{w}{v} \right)^{1/3}}$$

3. (15 points) Consider the following utility function:

$$U(X, Y) = 2XY$$

- (a) Solve for the Marshallian demands g_x and g_y .
- (b) Income is $I = 4$ and prices are $p_x = \frac{2}{3}$ and $p_y = 1$. What are the optimal demands and level of indirect utility?
- (c) Suppose there is a tax $T = 2$ on X (new price = $p_x + T$). How much additional income would a consumer need to receive to be as well-off as before the tax?

$$(a) \quad g_x = \frac{1}{4+1} \frac{I}{p_x} = \frac{I}{2p_x}$$

$$g_y = \frac{I}{2p_y}$$

$$(b) \quad g_x = \frac{4}{4/3} = 3$$

$$g_y = \frac{4}{2} = 2$$

$$V = 2(3)(2) = 12$$

$$(c) \quad V = 2 \frac{I^2}{(2p_x)(2p_y)} = \frac{I^2}{2p_x p_y}$$

$$I = (2p_x p_y V)^{1/2}$$

$$p_x^{new} = \frac{2}{3} + 2 = \frac{8}{3}$$

$$V^{old} = 12 \Rightarrow I^{new} = \left(2\left(\frac{8}{3}\right)(1)(12) \right)^{1/2} = (64)^{1/2} = 8$$

$$I^{new} - I^{old} = 8 - 4 = \boxed{4}$$

Bonus: What is the correct spelling of your Econ 11 Professor's last name?

Mazzocco